

PAPER_381 — SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law

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Section: SGR1745 compressed MUGE computation — full 8-term breakdown

Session: 104 (Complete Re-Analysis — individual term values undiscovered in PAPER_372)

CP4 Class: SGR1745CompressedMUGESpectralTermDecompositionCalculator (CP4 #32)

Abstract

This paper presents a UQFF analysis of SGR1745 Compressed MUGE Spectral Term Decomposition & Perturbation Dominance Law, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_372 documented the final compressed MUGE result for SGR1745 ($g \approx 1.782e39 \text{ m/s}^2$) and established the 8-function modular structure. However, it did NOT record the individual magnitudes of all 8 terms side-by-side. This paper fills that gap with the first complete **spectral term decomposition** showing the relative magnitude of each compressed MUGE contribution.

The key discovery: the **perturbation term dominates by 27 orders of magnitude** over the Newtonian base, revealing why the compressed model is **unphysical at magnetar scale**.

2. SGR1745 System Parameters

Parameter	Symbol	Value	Units
Mass	M	2.984e30	kg
Radius	r	1×10^4	m
Magnetic field	B	1×10^{10}	T
Critical B-field	B_crit	1×10^{11}	T
Age	t	3.799e10	s
Redshift	z	0.0009	—
Expansion velocity	v_exp	1×10^3	m/s
Dark matter mass	M_DM	1×10^{28}	kg
Density contrast	$\delta\rho/\rho$	0.1	—

3. Complete 8-Term Spectral Decomposition

Term 1: Newtonian Base

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{(10^4)^2}$$

$$g_{\text{base}} = 1.991 \times 10^{12} \text{ m/s}^2$$

Term 2: Hubble Expansion Correction

$$g_{\text{expansion}} = g_{\text{base}} \times (1 + H_0 t)$$

At $t = 3.799 \times 10^{10}$ s and $H_0 = 2.269 \times 10^{-18}$ s⁻¹:

$$1 + H_0 t = 1 + (2.269 \times 10^{-18})(3.799 \times 10^{10}) = 1.0000000862$$

$g_{\text{expansion factor}} = 1.0000000862$ (negligible at magnetar age)

Term 3: Superconducting Correction (Meissner linear)

$$g_{\text{SC}} = g_{\text{base}} \times (1 + H_0 t) \times \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

$$\left(1 - \frac{10^{10}}{10^{11}}\right) = 0.9$$

$g_{\text{SC adj}} = 1.792 \times 10^{12}$ m/s ²

Term 4: External BH Gravity (Ug3')

$$U'_{g3} = \frac{GM_{\text{BH}}}{r_{\text{BH}}^2} \quad (\text{external Sgr A* at } r = 26 \text{ kpc})$$

$U'_{g3} = 6.746 \times 10^{-5}$ m/s ²

Term 5: Cosmological Constant Floor

$$g_{\Lambda} = \frac{\Lambda c^2}{3} = \frac{1.1 \times 10^{-52} \times (3 \times 10^8)^2}{3}$$

$g_{\Lambda} = 3.3 \times 10^{-36}$ m/s ² (effectively zero at magnetar scale)

Term 6: Quantum Coherence Term

$$g_{\text{quantum}} = \frac{\hbar}{\Delta x \cdot \Delta p} \cdot \int \psi^\dagger \hat{H} \psi dV \cdot \frac{2\pi}{t_{\text{Hubble}}}$$

With $\Delta x \cdot \Delta p = 10^{-68}$ J²·s² and coherence integral = 2.176×10^{-18} J:

$g_{\text{quantum}} = 3.316 \times 10^{-35}$ m/s ²

Term 7: Fluid Coupling

$$g_{\text{fluid}} = \rho_{\text{fluid}} \cdot V_{\text{sys}} \cdot g_{\text{local}}$$

$g_{\text{fluid}} = 4.189 \times 10^{-2}$ m/s ²
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Term 8: Dark Matter Perturbation (DOMINANT TERM)

$$g_{\text{pert}} = (M + M_{\text{DM}}) \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3} \right)$$

At $r = 10^4$ m:

$$\frac{3GM}{r^3} = \frac{3 \times 6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{(10^4)^3} = 6.0 \times 10^{10}$$

$$(M + M_{\text{DM}}) \approx (2.984 \times 10^{30} + 10^{28}) \approx 3.0 \times 10^{30} \text{ kg}$$

$$g_{\text{pert}} = 1.782 \times 10^{39} \text{ m/s}^2 \quad (\text{DOMINANT} - 27 \text{ orders above base})$$

4. Complete Spectral Decomposition Table

Term	Formula	Value (m/s ²)	Orders above base
Base (Newtonian)	GM/r ²	1.991e12	—
SC adj (×0.9)	×(1−B/B_crit)	1.792e12	0
Ug3' (ext. BH)	GM_BH/r_BH ²	6.746e-5	−17
Cosmological floor	Λc ² /3	3.3e-36	−48
Quantum coherence	ħ⟨Ĥ⟩·2π/t_H	3.316e-35	−47
Fluid coupling	ρ_f·V·g_loc	4.189e-2	−14
Perturbation (DM)	(M+M_DM)(δρ/ρ+3GM/r³)	1.782e39	+27

5. Perturbation Dominance Law

Statement: For compact objects at $r \sim 10^4$ m (magnetar scale), the dark matter perturbation term in the Compressed MUGE dominates by ≥ 27 orders of magnitude over the Newtonian base.

Physical origin: The $3GM/r^3$ factor scales as r^{-3} — making it catastrophically large at magnetar radii:

$$\left. \frac{3GM}{r^3} \right|_{r=10^4} = 6.0 \times 10^{10} \gg \left. \frac{3GM}{r^3} \right|_{r=1 \text{ AU}} \approx 1.7 \times 10^{-23}$$

Implication: The Compressed MUGE formulation is **unphysical** for compact objects. At magnetar scale, $r = 10^4$ m violates the assumption that $(M + M_{DM}) \cdot \frac{3GM}{r^3}$ remains a

small correction. The **Resonance MUGE model** (PAPER_371) with fluid-dominant term $a_{fluid_freq} = 1.773 \times 10^{-9} \text{ m/s}^2$ is the physically appropriate description.

Validity domain criterion:

$$\text{Compressed MUGE valid when: } \frac{3GM}{r^3} \ll \frac{\delta\rho}{\rho} \Rightarrow r \gg \left(\frac{3GM}{\delta\rho/\rho} \right)^{1/3}$$

For SGR1745 with $\delta\rho/\rho = 0.1$:

$$r_{\min_compressed} = \left(\frac{3 \times 6.674 \times 10^{-11} \times 2.984 \times 10^{30}}{0.1} \right)^{1/3} \approx 1.3 \times 10^7 \text{ m}$$

The magnetar radius $r = 10^4 \text{ m}$ violates this by 3 orders — confirming Compressed MUGE is invalid.

6. Connection to PAPER_379 Dual-Model Comparison

PAPER_379 showed the **48-order divergence** between compressed ($1.782 \times 10^{39} \text{ m/s}^2$) and resonance ($1.773 \times 10^{-9} \text{ m/s}^2$) results for SGR1745. This paper explains the **root cause**: the perturbation term alone produces a 27-order excess, and the two models diverge by 48 orders because compressed MUGE sums a physically explosive term that simply should not apply at $r = 10^4 \text{ m}$.

Model selection rule (first formal statement): - $r < 10^7 \text{ m}$ (compact stars, NS, BH): Use **Resonance MUGE** - $r > 10^{10} \text{ m}$ (galaxies, clusters, cosmological): Use **Compressed MUGE** - Transition zone: Both models compared, resonance preferred when fluid term dominates

7. Key Equations

$$g_{\text{compressed}}^{\text{SGR1745}} = \underbrace{1.792 \times 10^{12}}_{\text{Newt+SC}} + \underbrace{6.746 \times 10^{-5}}_{\text{Ug3'}} + \underbrace{4.189 \times 10^{-2}}_{\text{fluid}} + \underbrace{1.782 \times 10^{39}}_{\text{perturbation}}$$

$$\approx 1.782 \times 10^{39} \text{ m/s}^2 \quad (\text{perturbation-dominated})$$

Resonance MUGE (physically correct for this system):

$$g_{\text{resonance}}^{\text{SGR1745}} \approx a_{\text{fluid_freq}} = 1.773 \times 10^{-9} \text{ m/s}^2$$

Full divergence: $1.782 \times 10^{39} / 1.773 \times 10^{-9} \approx 10^{48}$ — 48 orders of magnitude.

8. References Within Codebase

- PAPER_372: Compressed UQFF B/Bcrit Superconductivity — framework and final value
- PAPER_371: MUGE 12-Term Resonance — correct model for SGR1745
- PAPER_379: Dual-model 7-system comparison — the 48-order divergence

- CompressedUQFFBcritSuperconductivityCalculator (CP4 #15): Full 8-function implementation

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